

CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY

1st Semester of B. Pharm. Examination

University Theory Examination January 2018

BPH1001 Human Anatomy & Physiology-I

Date: 01.01.2018, Monday

Time: 10.00 a.m. to 1.00 p.m.

Maximum Marks: 75

Instructions:

1. There are three sections in this question paper.
2. SECTION – I comprises of Question 1. Total marks for Section 1 are 20. There are 20 sub-questions (MCQ type). Answers to SECTION – I are to be given in Answer Sheet for MCQ type questions provided to you. Maximum time allotted for SECTION – I is 30 minutes. Answers to SECTION – I must be written during the first 30 minutes of the examination.
3. Answers to SECTION – II and SECTION – III are to be provided in separate Main Answer Books provided to you.
4. Figures to right indicate marks.
5. Draw neat labeled sketches wherever necessary.

SECTION - I

Q 1 Attempt all questions. Each question is of one mark.

20

1. During pulmonary circulation blood leaves the
[A] right ventricle and goes directly to the aorta
[B] right atrium and goes directly to the lungs
[C] right ventricle and moves to the lungs
[D] right atrium and goes directly to the left ventricle
2. Hypoxia stimulates the kidneys to step up the release of _____:
[A] Thrombopoietin
[B] Erythropoietin
[C] Hemoglobin
[D] Globin
3. The activity of which contractile protein is altered to regulate smooth muscle contraction?
[A] Actin
[B] Myosin
[C] Calmodulin
[D] Tropomyosin
4. Which intercellular junctions directly allow the passage of small molecules and ions between the cytosol of one cell and its neighbour without movement into interstitial fluid?
[A] Gap junctions
[B] Focal adhesions
[C] Zonula occludens
[D] Desmosome
5. The following are examples of connective tissue, except:
[A] Bones
[B] Tendons
[C] Skeletal muscle
[D] Cartilage

6. Following are the examples of active transport except.
 - [A] Receptor mediated endocytosis
 - [B] Phagocytosis
 - [C] Osmosis
 - [D] Pinocytosis
7. The main components of the skeletal system are:
 - [A] Blood, bones, cartilage and tendons
 - [B] Bones, cartilage, joints and tendons
 - [C] Bones, cartilage, joints and ligaments
 - [D] Cartilage, joints, ligaments and tendons
8. Skin colour is determined by which pigment?
 - [A] Melanin
 - [B] Protein
 - [C] Anthocyanin
 - [D] Gluten
9. _____ of ECG represents ventricular repolarization.
 - [A] P wave
 - [B] QRS complex
 - [C] S wave
 - [D] T wave
10. The following are examples of the function of nervous tissue, except:
 - [A] Exhibits sensitivity to different stimuli
 - [B] Converts stimuli into nerve impulses
 - [C] Strengthens nerve impulses
 - [D] Conducts nerve impulses to other neurons
11. Blood moving from the atria into the ventricles flow through which two valves?
 - [A] Pulmonary and mitral (bicuspid or left AV)
 - [B] Aortic and pulmonary
 - [C] Tricuspid (right AV) and mitral (bicuspid or left AV)
 - [D] Tricuspid (right AV) and aortic
12. Blood from the heart flows through the blood vessels in which order?
 - [A] Veins, arterioles, capillaries, venules, arteries
 - [B] Capillaries, arterioles, arteries, venules, veins
 - [C] Arteries, arterioles, capillaries, venules, veins
 - [D] Arteries, capillaries, arterioles, venules, veins
13. What is the large vessel running parallel to the spinal column transporting lymph called?
 - [A] The aorta
 - [B] The lymphatic duct
 - [C] The thoracic duct
 - [D] The right subclavian vein
14. Antibodies are formed from which white blood cell?
 - [A] Neutrophil
 - [B] Basophil
 - [C] B-lymphocyte
 - [D] Monocyte

15. During mitosis, identical copies of DNA molecules organize into chromatid pairs within the chromosome structure. These pairs are connected to each other at the chromosome's centromere. This phase is called:
 [A] Prometaphase
 [B] Metaphase
 [C] Prophase
 [D] Telephase
16. Which is an example of an irregular bone?
 [A] Vertebra
 [B] Patella
 [C] Scapula
 [D] Metacarpal
17. Which of the following would not be considered a fibrous joint?
 [A] The joint between the parietal bone and the occipital bone
 [B] The joint between a tooth and the mandible
 [C] The pubic symphysis
 [D] The interosseous membrane of the leg
18. The layer of the heart wall primarily responsible for the heart's pumping action is the:
 [A] Myocardium
 [B] Endocardium
 [C] Epicardium
 [D] Pericardium
19. The volume of blood, in liters that each ventricle of the heart ejects every minute is Known as:
 [A] Stroke volume
 [B] Cardiac output
 [C] Heart rate
 [D] Blood pressure
20. Neutrophils perform which of the following functions?
 [A] Produce antibodies
 [B] Phagocytize bacteria
 [C] Destroy infected body cells
 [D] Deliver carbon dioxide to the lungs

SECTION - II

- Q 2 Draw neat and clean sketch of Gross anatomy of Human Heart. Explain the role of heart valves in cardiac conduction system. 10
- Q 3 Attempt any **TWO** of the followings:
- A Draw neat and clean diagram of Eye with mentioning important functions of each part. 05
- B Enlist the Cranial and spinal nerves. 05
- C How Peripheral nervous system is involved in control of cardiovascular system? Classify Peripheral Nervous system in short. 05

SECTION – III

- Q 4** Attempt any **ONE** of the following
- A** Discuss composition and functions of blood. Discuss the life cycle of RBC. 10
- B** Discuss the followings in brief: 10
- i) Draw a labeled diagram of neuromuscular junction. Give physiology of muscle contraction in short.
- ii) Explain about lymph circulation and functions of lymphatic system.
- Q 5** Attempt any **FIVE** of the following
- A** Explain structure and functions of skin with diagram. 05
- B** Write a short note on feedback system. 05
- C** Compare and contrast of Muscular and nervous tissue. 05
- D** Classify the joints with example. Write a note on synovial joints. 05
- E** What is hemostasis? Write a note on blood clotting in detail. 05
- F** Discuss the different transport mechanisms across the cell membrane. 05